Ferroelectricity in Hafnium Oxide Thin Films

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Ferroelectricity in hafnium oxide thin films

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We report that crystalline phases with ferroelectric behavior can be formed in thin films of SiO_2 doped hafnium oxide. Films with a thickness of 10 nm and with less than 4 mol. % of SiO_2 crystallize in a monoclinic/tetragonal phase mixture. We observed that the formation of the monoclinic phase is inhibited if crystallization occurs under mechanical encapsulation and an orthorhombic phase is obtained. This phase shows a distinct piezoelectric response, while polarization measurements exhibit a remanent polarization above 10 μC/cm^2 at a coercive field of 1 MV/cm, suggesting that this phase is ferroelectric. Ferroelectric hafnium oxide is ideally suited for ferroelectric field effect transistors and capacitors due to its excellent compatibility to silicon technology. © *2011 American Institute of Physics*. [doi:10.1063/1.3634052]

Hafnium oxide is known to exist in three different crystal phases at normal pressure: A monoclinic phase at room temperature, a tetragonal phase above 2050 K, and finally a cubic phase above 2803 K.[1] The stable region of the tetragonal phase extends to lower temperatures in nano-scale crystallites due to the surface energy effect.[2] As a consequence, the crystallization in thin films tends to proceed by nucleation in a tetragonal phase and a martensitic transformation to the monoclinic phase during crystal growth. This phase transformation involves volume expansion and shearing of the unit cell. The admixture of sufficient SiO_2 (between 5 and 10 mol. %) has been found to stabilize the tetragonal phase in HfO_2.[3–5] In addition, it was reported that also the presence of a top electrode during crystallization of HfO_2 thin films leads to a reduction of the monoclinic phase fraction[6] and a significant increase in the dielectric constant.[7] We investigated the influence of mechanical encapsulation (capping) on Si:HfO_2 thin films at lower Si content where the complete stabilization of the tetragonal phase does not yet occur.

Si:HfO_2 was deposited by a metal organic atomic layer deposition process based on Tetrakis-(ethylmethylamino)-hafnium (TEMA-Hf), Tetrakis-dimethylamino-silane (4DMAS), metalorganic precursors, and ozone. The silicon content was defined by varying the cycle ratio of the precursors and monitored by secondary ions mass spectrometry and elastic recoil detection analysis on samples without thermal treatment. The crystallization temperature of all films in the used thickness (7-10 nm) and composition range (2.5–6 mol. % SiO_2) was above 500 °C. Titanium nitride (TiN) electrodes were deposited by a chemical vapor deposition process based on $TiCl_4$ and NH_3. Crystallization of the Si:HfO_2 thin films was induced by a 1000 °C/20 s anneal after top electrode deposition, unless noted otherwise. All measurements were performed on metal-insulator-metal (MIM) capacitors and on blanket wafers for physical measurements.

Grazing-incidence x-ray diffractograms (GI-XRD) of Si:HfO_2 thin films were measured on a Brucker D8 Discover. Polarization hystereses were characterized using an aixACCT TF Analyzer 2000 system at a frequency of 1000 Hz. Piezoelectric displacement measurements were made using an aixACCT systems double beam laser interferometer. The measurements were conducted at an applied frequency of 1234 Hz and averaged over 1000 measurements. The small signal capacitance-voltage characteristics of the MIM capacitors were determined with an Agilent 4284 LCR meter. Typical measurements conditions were 10 kHz, 25 mV Vpp on a device with an area of 10^{-4} cm^2.

By lowering the deposition temperature of the TiN top electrode below the crystallization temperature of the Si:HfO_2 dielectric, it was possible to encapsulate amorphous films and induce the crystallization in a subsequent thermal treatment step above the crystallization temperature. We found that capacitors manufactured with this scheme exhibited significant non-linearity in their capacitance-voltage behavior, which did not occur if crystallization was induced prior to capping (Fig. 1). A comprehensive characterization was made by electrical polarization measurements on capped Si:HfO_2 films at different compositions. A gradual transition from ferroelectric to antiferroelectric polarization curves is observed with increasing silicon content, as in Fig. 2. The maxima in the corresponding AC C-V curves precede

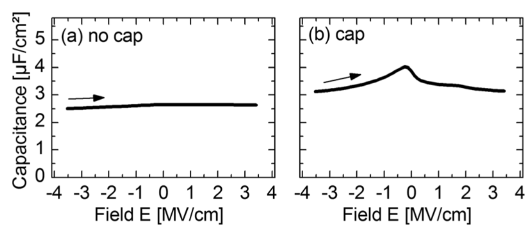

FIG. 1. (a) AC C-V characteristics of a 8.5 nm Si:HfO_2-MIM capacitor with ∼3 mol. % SiO_2 doping where crystallization was induced by a 900 °C/20 s anneal before deposition of the top electrode. An additional 1000 °C/20 s anneal was applied after top electrode deposition. (b) AC C-V characteristics of the same film stack where crystallization was induced after top electrode deposition by a 1000 °C/20 s anneal ("capped" crystallization).

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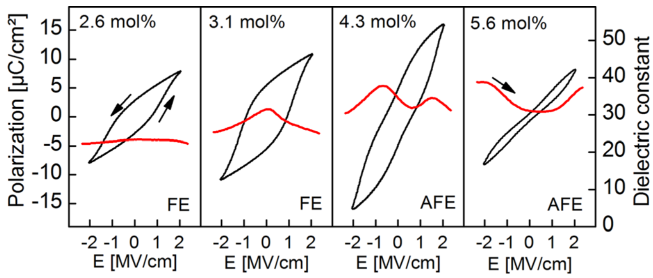

FIG. 2. (Color online) Polarization hystereses (black) and AC C-V characteristics (thick red) of a set of "capped" MIM capacitors with 8.5 nm insulator thickness. The shape of the polarization loops changes from ferroelectric to antiferroelectric with increased SiO_2 admixture. This is reflected in one or two peaks in the AC C-V sweeps, respectively.

microstructural changes in the thin films and represent a reversible component of the total polarization.[8] Further evidence for ferroelectric and antiferroelectric behavior was obtained by measurement of the piezoelectric displacement[9] (Fig. 3). The mechanical response of the ferroelectric sample shows clear evidence of piezoelectric behavior, which is a prerequisite for ferroelectricity.[10] A symmetric, electrostrictive response is found for the anti-ferroelectric sample, which correctly indicates a centrosymmetric crystalline phase. This observation confirms microstructural transformations in the bulk and rules out parasitic electronic effects, such as charge trapping,[11] as the origin of the observed polarization loops.

Ferroelectricity is intimately related to the crystal structure; only non-centrosymmetric space groups can exhibit piezoelectric and ferroelectric properties.[10] All common and reported HfO_2 bulk phases (space groups $p4_2/nmc$, $Fm\bar{3}m$, $P2_1/c$, Pbca, Pbcm) are centrosymmetric and, therefore, not ferroelectric. We performed GI-XRD measurements of thin film samples at a composition which showed ferroelectricity. As shown in Fig. 4, distinct differences between crystallization with and without capping are observed. The diffraction pattern of the uncapped sample shows a good match to the

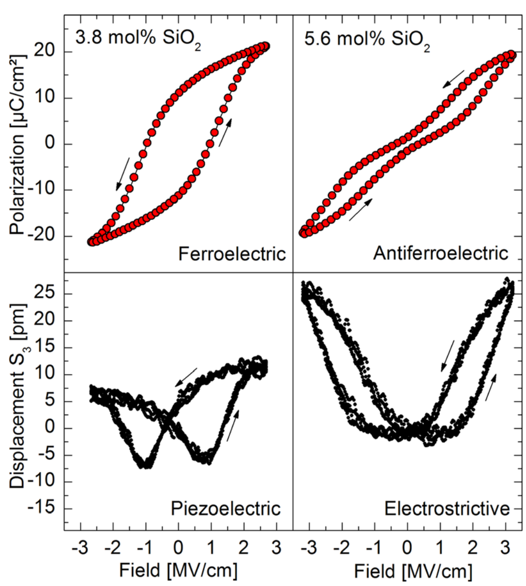

FIG. 3. (Color online) Polarization (top row) and piezoelectric displacement (bottom row) measurement of MIM capacitor samples with ferroelectric (left column) and antiferroelectric (right column) $Si:HfO_2$ insulators.

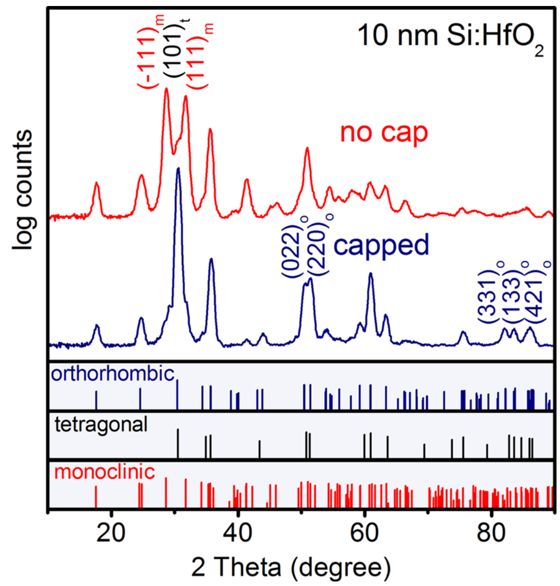

FIG. 4. (Color online) Grazing incidence x-ray diffraction measurements of two $Si:HfO_2$ samples of the same composition where crystallization was induced with and without capping. The molar SiO_2 content is approximately 3%, a composition where ferroelectricity was observed in MIM capacitors. The reflex positions for the $Pbc2_1$ phase were calculated based on o-ZrO_2 literature data, with isotropic rescaling by a factor of 0.992 to accommodate for substitution of smaller silicon atoms.

monoclinic phase, with a smaller fraction of tetragonal crystallites, as evident from the $(101)_t$ reflection of the tetragonal phase and the $(11\bar{1};)_m$ and $(111)_m$ reflections of the monoclinic phase. No $(101)_t$ splitting is observed for the capped sample. This suggests that the formed phase has an orthogonal unit cell and no monoclinic phase was formed. However, a number of reflections that are not common to the tetragonal phase, notably a triplet at 83°, indicate the occurrence of a lower symmetry phase.

The observed diffraction pattern can be well matched to an orthorhombic phase. The known orthorhombic forms of HfO_2 (space groups Pbca and Pbcm) are centrosymmetric and are, therefore, not ferroelectric. However, a similar diffraction pattern has been reported for a rare orthorhombic phase with space group $Pbc2_1$ in Mg stabilized ZrO_2.[12–14] Since the crystal chemistry of ZrO_2 is almost identical to that of HfO_2, the same crystalline phases are expected for both compounds. In contrast to the other two orthorhombic forms of ZrO_2, the $Pbc2_1$ space group is not centrosymmetric and, therefore, does not exclude ferroelectricity. Curiously, it seems that the dielectric properties of this phase in ZrO_2 have never been investigated.

It has been shown that the $Pbc2_1$ phase emerges in $Mg:ZrO_2$ during cooling[15] through a transformation from tetragonal crystallites enclosed in the $Mg:ZrO_2$ matrix. Usually, the tetragonal phase transforms into the monoclinic phase under these conditions. It has been suggested that the t→m transformation is inhibited by external stress[14] and a shearless transformation into orthorhombic ZrO_2 takes place instead.

Very similar conditions exist during the crystallization of capped $Si:HfO_2$ thin films. Crystallization is induced by high temperature annealing. Due to Ostwald's rule and increased stability at high temperatures, the initial nucleation

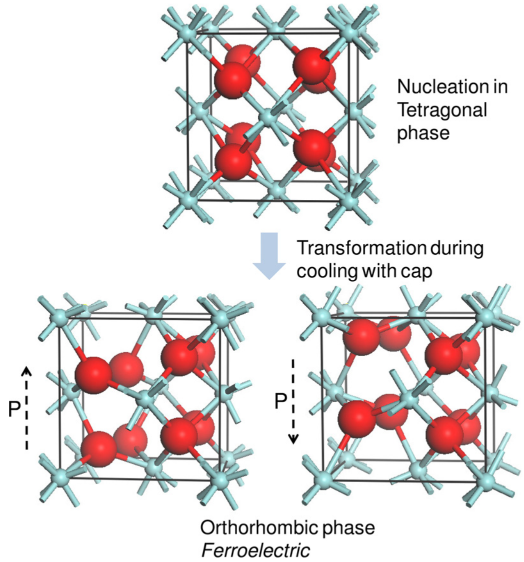

FIG. 5. (Color online) The formation of the orthorhombic phase proceeds by transformation from the tetragonal phase during cooling with a cap. The bottom row indicates two different polarization states of the ferroelectric phase. (Large atoms are oxygen and small atoms are hafnium).

is believed to take place in a metastable tetragonal phase. If the film is not capped, a partial transformation into the monoclinic phase occurs. In the presence of a cap, the shearing of the unit cell is mechanically inhibited and a transformation into the orthorhombic HfO_2 $Pbc2_1$ phase may take place (Fig. 5).

The exact origin of the *antiferroelectric* behavior is not clear. The potentially antiferroelectric nature of a metastable tetragonal phase in ZrO_2 was suggested earlier.[13] This form is isostructural to the high temperature tetragonal phase, but with a lower c/a ratio close to 1, and has also been reported in doped HfO_2.[16] The double hysteresis can be explained by field induced phase transformations close to the ferroelectric transition, as observed in relaxor ferroelectrics.[17,18]

A major role of SiO_2 in this system is to increase the crystallization temperature of Si:HfO_2, favoring a controlled crystallization *after capping*. A secondary role of SiO_2 is reducing the stability of the monoclinic phase.[4] Considering the low level of doping, SiO_2 is not an active component of the ferroelectric phase. Due to the similarity to HfO_2, we expect that ferroelectric behavior occurs in ZrO_2 based thin films as well.

In conclusion, we have reported evidence for the formation of ferroelectric and antiferroelectric crystalline phases in SiO_2 doped HfO_2 thin films. Based on XRD measurements and analogies to Mg:ZrO_2, it was argued that the ferroelectric phase is orthorhombic with a $Pbc2_1$ space group. The phase is formed due to inhibition of the t→m transformation by mechanical confinement. The occurrence of ferroelectricity in Si:HfO_2 is remarkable as it represents one of very few metal oxides which are thermodynamically stable on silicon,[19] potentially enabling a number of device concepts relying on silicon/ferroelectric junctions.[20] The high valence band offset of HfO_2 to silicon allows low leakage devices, unlike other materials as $SrTiO_3$.[21]

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